

Unit 3, Lesson 6: Solving Real-World Problems with Rational Numbers – Part 2

Benchmark

MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.

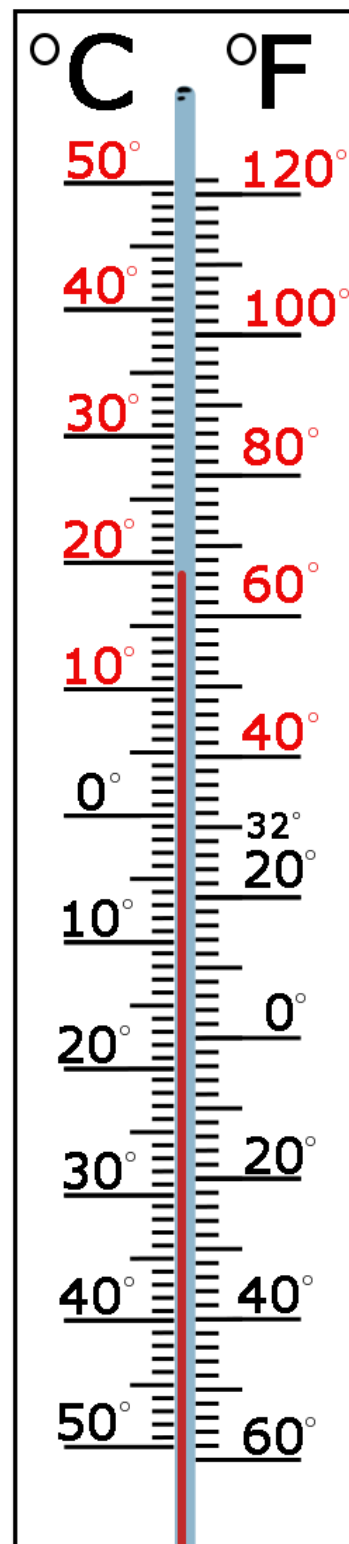
Learning Target

- ☐ I can solve problems with one or two operations by multiplying and/or dividing rational numbers.

3.6.1 Warm-Up: Math Fail

A picture of a thermometer is shown.

1. Explain the error in the image.
2. Describe how to fix the image so that there is no "math fail."

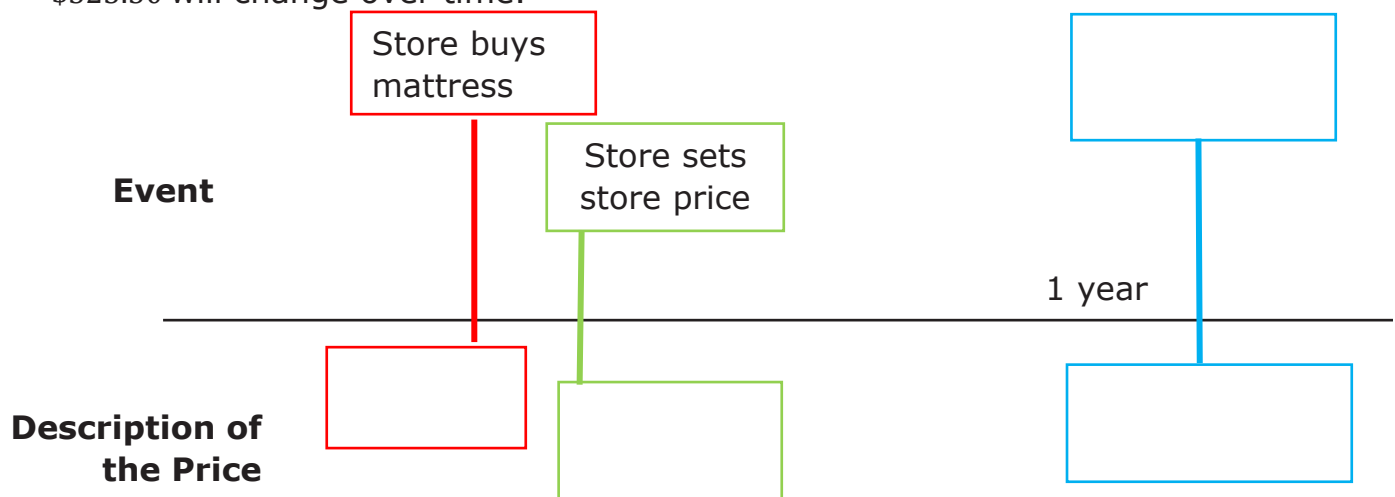


3.6.2 Collaboration: Solving a Problem with Rational Numbers

In order to make a profit, stores sell products for more than what they paid for them. For example, a local furniture store buys a mattress from a manufacturer for a certain price. The store then sets the store price at $2\frac{1}{2}$ times what they paid the manufacturer for it. If the mattress hasn't sold in a year, then the store will reduce its price by $\frac{1}{3}$ of the original store price.

Work with your partner to complete the following based on the information given.

- Drawing a diagram can help make sense of the problem. Without calculating, complete the timeline to show how the price of a mattress the store purchases for \$325.50 will change over time.

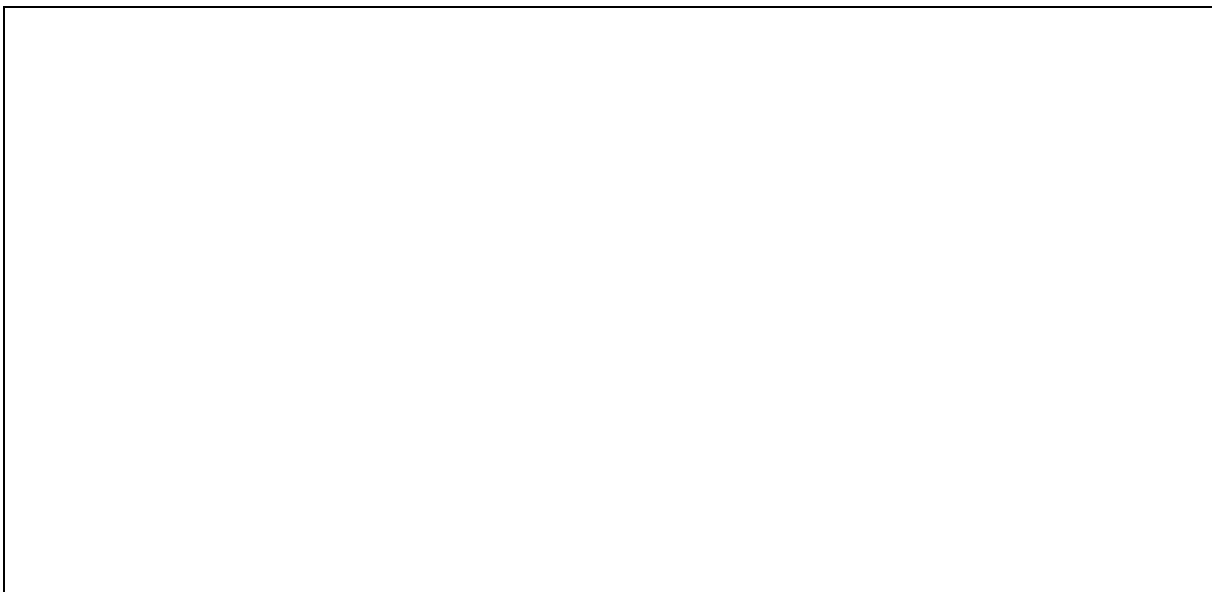


- Explain whether the store's first price will be greater than or less than the price they paid the manufacturer.
- What is the first price the store will set for the mattress?
- Explain whether the price after a year will be greater than or less than the store's first price.
- What is the sale price of the mattress after one year?

6. If the store sells the mattress after one year, explain if the store will make or lose money on the mattress.

3.6.3 Guided Instruction: Solving Real-World Problems with Rational Numbers

1. Ramon paid \$18,360.00 for a new car. By the end of the first year, the value of the car changed by $-\frac{3}{14}$ of its original value. By the end of the second year, the value of the car changed by $-\frac{2}{9}$ of its value at the end of the first year.
 - a. Is the value of the car increasing or decreasing during these two years?
 - b. By what amount did the value of the car change at the end of the first year?
 - c. What is the value of Ramon's car at the end of the second year?
2. A farmer is building a fence around a space on his property. The space has five equal sides. The farmer determined that he should buy 138.75 yards (yd) of fencing. The fence costs \$133.21 per yd.
 - a. Draw and label a diagram to make sense of the problem.



- b. How much fencing is used for each side?
 - c. What is the cost for fencing on one side of the space?
3. A water well drilling rig dug to a depth of -147.5 feet (ft). The rig drills $-2\frac{1}{2}$ ft in one hour.
- a. How many hours has the drill been running?
 - b. At this rate, how many hours will it take until the drill reaches a depth of -250 ft?
4. A marine biologist was doing research in a one-person submarine. The submarine started to ascend (go up) from a position of $-130\frac{1}{2}$ feet. The submarine surfaces (goes up) at a rate of 23.3 feet per minute. How long, in minutes, will it take the submarine to reach the surface?

3.6.4 Your Turn!

1. Delaney wants to sell her bicycle. The original price of the bike was \$150. The value of the bike has changed by $-\frac{2}{5}$ of the original price. What is the current value of the bike?
2. At the beginning of the day on Monday, Stephon had \$15.78 in his lunch account. At the end of the day on Friday, he had \$1.53 in his lunch account. If Stephon purchased the same lunch each day on Monday through Friday, describe how the account changed each day.

3.6.5 Wrap-Up: Ticket Out the Door

1. If the temperature has been changing at a rate of -3.64°F every hour for $3\frac{1}{2}$ hours, how much has the temperature changed over that time?